

Netflix → show a new movie to 20000 users
 ↳ exactly predict who will watch it? (No)
 ↳ predict approximately how many people will watch it?
probability theory
 max predict $\Delta 8\%$ → right zone
 Spam detection → probability spam = 92%
 Loan default → " default = 18%
 disease prediction → " disease = 78%
 Customer churn → " churn = 35%

future outcomes are random
 ↳ randomness is modeled using Random variables.

→ A variable whose value depends on outcome of a random experiment.

No. of orders tomorrow → 500, 1000, 1200, 450
 X = Number of orders tomorrow
 ↳ is uncertain
 ↳ random variable. future ----- Unknown ----- R.V. (X)

X = customer purchase
 ↳ 0 = no purchase
 ↳ 1 = purchase

Random variable

① Discrete Random variable
 ↳ countable values

② Continuous random variable
 Not countable
 (CLV)
 Customer Lifetime Value
 $I \rightarrow 15250.50$
 $II \rightarrow 15250.58$ } infinite precision possible

PMF (Probability Mass Function)
 ↳ dice 1, 2, 3, 4, 5, 6
 ↳ value → probability
 $P(X=x)$
 Probability that random variable (X) takes value x

discrete outcomes.
 ↳ value → probability
 $P(X=x)$
 Probability that random variable (X) takes value x

PMF	Probabilities
X	x
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

PMF → random → every probability ≥ 0
 ↳ random → all probabilities sum = 1

CDF → Cumulative distribution Function

1	1/6	probability	→ 1/6
2	2/6	$F(x) = P(X \leq x)$	Day 1 → 10 10
3	3/6	↓	Day 2 → 20 30
4	4/6	↓	Day 3 → 15 45
5	5/6	↓	
6	6/6	↓	

CDF = probability of every value x

Marks	Probability
40	0.2
50	0.3
60	0.1
70	0.4

CDF

probability score ≤ 60 ? → 0.2 + 0.3 + 0.1 = 0.6

* roll a die → 1M times.

↳ Expected value.
 ↳ center of gravity of distribution.
 ↳ Average outcome over infinite repetitions.

$E[X] = \sum x_i \cdot P(x_i)$
 Take each value multiply by its probability add anything

Value	Probability
0	0.1
30	0.3
50	0.4
100	0.2

expected score → $0 \times 0.1 + 30 \times 0.3 + 50 \times 0.4 + 100 \times 0.2$
 $9 + 20 + 20 = 49$

Variance

A	B
49	0
49	100
49	0
49	100
49	49

 ↳ avg = 49
 ↳ Spread
 ↳ variability
 ↳ uncertainty

Variance → how far observations spread from average.
 Sample 30
 32
 33
 35
 24
 28
 26
 Mean 16
 14
 15
 16
 90

Model A
 1
 2
 1
 2
 1
 1
 2
 5
 1.4
 Variance
 Model B
 0
 15
 0
 20
 0
 35
 5
 7

$Var(X) = E[X - E[X]]^2$
 → $\frac{1+2+3+1+5}{5} = 3$

Value	Mean	Distance	Sum
1	3	-2	4
2	3	-1	1
3	3	0	0
4	3	1	1
5	3	2	4

 ↳ $\frac{10}{5} = 2$

Standard deviation (σ) = $\sqrt{Var(X)}$ = $\sqrt{2}$ = 1.414
 Mayahoge → avg spend → 10000
 S.d. → 500
 $V = (S.d)^2$
 $= (500)^2 = 250000$
 avg. customer spend = 10,000 ✓
 S.d. = 500

mean	variance	std dev
10000	250000	500
10000	4000000	2000

Properties of Expectation.

P1 → $E(c) = c$
 P2 → $E[aX] = aE[X]$
 avg salary = 50000
 avg doubled.
 P3 → $E(X+Y) = E[X] + E[Y]$
 food cost → 300
 travel → 200
 expense total → 500
 Linearity of expectation

Bernoulli Distribution

Spam detection → Spam
 ↳ not spam

Credit card approval → Approve
 ↳ reject

Binary Problems

↳ B.D. → one trial
 ↳ only two outcomes: success → p
 ↳ failure → 1-p
 toss a coin → H → p = 0.5
 ↳ T → p = 0.5 → 1-p

PMF of Bernoulli

X	Probab.
0	1-p
1	p

$E[X] = p$

Average success rate equals probability of success.